



What are the

Assumptions of Kinetic Theory

1. The gas consists of very small particles, all with non-zero mass.
2. The number of molecules is large such that statistical treatment can be applied.
3. These molecules are in constant, random motion. The rapidly moving particles constantly collide with the walls of the container.
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4. The collisions of gas particles with the walls of the container holding them are perfectly elastic.
5. Except during collision the interactions among molecules are negligible (they exert no forces on one another).
6. The total volume of the individual gas molecules added up is negligible compared to the volume of the container. This is equivalent to stating that the average distance separating the gas particles is large compared to their size.
7. The molecules are perfectly spherical in shape, and elastic in nature.
8. The average kinetic energy of the gas particles depends only on the temperature of the system.
9. Relativistic effects are negligible.
10. Quantum-mechanical effects are negligible. This means that the inter-particle distance is much larger than the thermal de Broglie wavelength and the molecules are treated as classical objects.
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11. The time during collision of molecule with the container's wall is negligible as comparable to the time between successive collisions.
12. The equations of motion of the molecules are time-reversible.

Maxwell-Boltzmann Distribution

This describes particle speeds in gases, where the particles do not constantly interact with each other but move freely between short collisions. It also describes the probability of the particle's speed (the magnitude of its velocity vector) being near a given value as a function of the temperature of the system, the mass of the particle and that speed value. This probability distribution is named after James Clerk Maxwell and Ludwig Boltzmann.
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The most probable speed of a molecule with a mass m is:

$$V_{\text{mas}} = \sqrt{\frac{3kT}{m}} \quad \text{where:}$$

k = Boltzmann constant
 $k = 1.38 \times 10^{-23} \text{ J/K}$

The absolute temperature is proportional to the square of the rms velocity:

$$T = \left(\frac{m}{3k} \right) V_{\text{rms}}^2$$

The mean translational kinetic energy of the molecule is proportional to the mean absolute temperature.

$$KE = \frac{1}{2} mv^2 = \frac{2}{3} kT$$

Brownian Movement

Names after the Scottish botanist Robert Brown, it is the seemingly random movement of particles suspended in a fluid (i.e. a liquid such as water or air) or the mathematical model used to describe such random movements, often called a particle theory



LAWS OF THERMODYNAMICS

FIRST LAW

(About the conservation of energy)

"The change in the internal energy of a closed thermodynamic system is equal to the sum of the amount of heat energy supplied to the system and the work done on the system"
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SECOND LAW

(About entropy)

"The total entropy of any isolated thermodynamic system tends to increase over time, approaching a maximum value."

Clausius statement:

"Heat generally cannot flow spontaneously from a material at lower temperature to a material at higher temperature."

Kelvin statement:

"It is impossible to convert heat completely into work in a cyclic process."

Caratheodory statement:

"In the neighborhood of any equilibrium state of thermodynamic system, there are equilibrium states that are adiabatically inaccessible."
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THIRD LAW

(About absolute zero temperature)

"The entropy of all systems and of all states of a system is zero at absolute zero."

Lewis and Randal version:

"If the entropy of each element in some (perfect) crystalline state be taken as zero at the absolute zero temperature, every substance has a finite positive entropy, but at the absolute zero of temperature, the entropy may become zero."

ZEROth LAW

(About thermodynamic equilibrium)

"If two thermodynamic systems are separately in thermal equilibrium with a third, they are also in thermal equilibrium with each other."



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